Lecture Notes — 1st Presentation

Modern Cloud Forensics: A PRISMA-Guided Systematic Survey

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Estimated speaking time: ~3.1 minutes (185 seconds across 10 slides)

Slide 1

Title *(~10 sec)*

Good [morning/afternoon]. This is my first RMTW presentation, covering my individual research project: a PRISMA-guided systematic survey of cloud forensics. Today I'll walk through the problem, my research questions, the methodology, and the literature review — the foundation the rest of my research builds on.

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The Problem *(~25 sec)*

Let's start with why this matters. Cloud forensics faces four core problems. First, investigators have no physical access — evidence lives inside VMs, containers, and logs that the cloud provider controls, not the investigator. Second, evidence disappears fast, because multi-tenancy and elastic scaling mean an instance can vanish before anyone captures its state. Third, legal complexity — data crosses borders, and privacy laws conflict with each other. And fourth, current solutions are fragmented: AI, blockchain, and automation each address a piece of the problem, but nothing ties them together into one coherent approach. That gap is what this survey addresses.

Slide 3

Research Questions *(~25 sec)*

To structure the investigation, I defined four research questions, each tied to one of the problems on the previous slide. RQ1 asks how we can achieve scalable, forensically sound evidence acquisition across SaaS, PaaS, and IaaS. RQ2 asks what mechanisms guarantee integrity, authenticity, and non-repudiation of evidence. RQ3 examines how privacy laws and jurisdiction issues affect the admissibility of evidence. And RQ4 asks how standardized, interoperable frameworks can improve cloud forensic investigations. These four questions drive every section of the survey that follows, including the taxonomy I'll present at the mid-term.

Slide 4

Methodology (PRISMA) *(~25 sec)*

My methodology follows PRISMA — a four-stage systematic review process. Stage one, Identification: I searched IEEE Xplore, ACM Digital Library, ScienceDirect, Springer Link, and Google Scholar. Stage two, Screening: I removed duplicates and studies unrelated to cloud forensics by title and abstract. Stage three, Eligibility: I assessed each remaining paper for methodological soundness, technical contribution, and legal relevance. Stage four, Inclusion: the final corpus was mapped directly onto RQ1 through RQ4. One methodological note — I chose a qualitative synthesis rather than a meta-analysis, because the sources are too heterogeneous in method, dataset, and purpose to combine statistically.

Slide 5

Literature Review *(~25 sec)*

The existing literature clusters into four recurring themes, which map directly onto the four research questions. Evidence Acquisition studies show that IaaS is the most mature environment for forensics, while SaaS remains the most dependent on cloud provider cooperation. Integrity and Trust studies focus heavily on blockchain-based provenance, cryptographic hashing, and trusted logging architectures. Jurisdiction and Law studies highlight unresolved tension between GDPR, mutual legal assistance treaties, and the US CLOUD Act. And Standardization studies point to ISO/IEC 27037 and 27043 and NIST guidance as the closest things we have to standards — but adoption by cloud providers is still voluntary and inconsistent.

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Notable Recent Work *(~25 sec)*

A handful of very recent papers shaped this survey's direction. NIST's SP 800-201, published in 2024, is the closest thing we currently have to a unified cloud forensic reference architecture. Two papers from 2025 — by Mukti and Hanif, and separately by Vucha and Kamarapu — apply AI directly to cloud-native evidence acquisition and anomaly detection. Chourasiya and colleagues combine LSTM, Transformer, and graph neural network models for log analysis, reporting over 98% anomaly-detection accuracy. And two of the newest papers, from 2026 — Dhital's team, and Ekwempu — show respectively that AWS-native monitoring tools miss transient attacker activity in short-lived instances, and map the legal fragmentation investigators face in cross-border cloud cases.

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What's Next *(~20 sec)*

This literature review sets up the rest of my research pipeline. Next, I'll decompose the field into a four-dimensional taxonomy — that's what I'll present at the mid-term, along with a comparative analysis of the selected studies by methodology, environment, and contribution. After that, I'll propose an innovative, explainable, standards-aligned acquisition framework based on the gaps this review has surfaced. And finally, I'll pilot that framework against real cloud-native scenarios and write up the complete research article for the final presentation.

Slide 8

References (1/2) *(~10 sec)*

These are the first ten of twenty references shown here, out of forty-eight total in the survey. This page covers the foundational and mid-period studies — from Alshabibi's 2024 systematic review, through Manral's 2019 ACM Computing Surveys paper, which remains one of the most cited foundational works in this field.

Slide 9

References (2/2) *(~10 sec)*

This second page covers the more recent additions — the 2024 through 2026 cloud-specific work, including the NIST standards, the newest AI-driven cloud forensics papers, and the legal and jurisdictional studies. The complete list of all forty-eight references is in the full survey paper, which I can share on request.

Slide 10

Thank You *(~10 sec)*

Thank you. That covers the problem, the research questions, the methodology, and the literature review that will anchor the rest of this project. I'm happy to take any questions. My next milestone is the taxonomy and comparative analysis, which I'll present at the mid-term.